scientific reports

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Vertical stratification of soil stoichiometric drivers across urban forest parks in Shenzhen, China

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The vertical distribution of soil nutrients and their stoichiometry is shaped by landscape-scale processes and soil-forming factors, but how these controls vary by depth remains unclear in urban forest parks. We examined this in six urban parks in Shenzhen, China, all below 300 m elevation and on similar parent material, by analyzing soil carbon (C), nitrogen (N), and phosphorus (P) stoichiometry across four depths (0–10, 10–20, 20–40, 40–60 cm). Higher N/P and C/P ratios in surface soils indicate potential P deficiency. Soil pH mainly regulates stoichiometry, likely by affecting P availability. Precipitation and elevation shaped nutrient patterns through leaching control and organic matter buildup. Notably, dominant factors shifted with depth: climate, topography, and human management drove patterns in surface layers (0–20 cm), while plant diversity and inherent soil properties were more important at 20–60 cm. This shift suggests greater biological and legacy effects in deeper, less disturbed soil. Our findings show that soil stoichiometry marks biogeochemical separation with depth and underscores the need to include deep-soil processes in nutrient cycling models to better predict ecosystem stability under environmental change.

Keywords C/N/P stoichiometry, Vertical stratification, Urban forest park soil, Soil acidification, Phosphorus deficiency

Soil properties regulate terrestrial ecosystem structure and function, primarily by controlling the availability and cycling of nutrients[1,2]. The contents and stoichiometric ratios of carbon (C), nitrogen (N), and phosphorus (P) are critical for soil fertility, nutrient limitations, and cycling efficiency. Understanding how environmental factors shape vertical soil C, N, and P stoichiometry is key for assessing urban forest park ecosystems. Nutrient cycling processes, shaped by plant uptake, litter decomposition, and microbial activity, depend on complex interactions among biotic and abiotic factors[3,4]. Spatial patterns of soil nutrients and their ratios are influenced by vegetation, land use, topography, climate, and human activities[5,6]. Disruptions to these patterns can degrade plant communities and ecosystem functions[7,8]. Thus, identifying the key mechanisms that regulate soil nutrient stoichiometry across soil depths is central to predicting urban forest park ecosystem responses to environmental change.

Soil depth is a fundamental control on nutrient distribution and cycling. In most terrestrial ecosystems, surface soils accumulate more organic matter and nutrients than deeper layers because they receive continuous inputs from plant litter, root exudates, and microbial necromass[9,10]. Additionally, surface soils have higher oxygen content, moisture, and microbial activity, which accelerate decomposition and nutrient turnover[11]. As a result, plant roots are concentrated near the surface to optimize water and nutrient uptake[12,13]. Although leaching and runoff can move mobile nutrients downward, surface layers typically retain higher total nutrient pools due to large organic inputs[14,15]. Consequently, nutrient concentrations and stoichiometric ratios often exhibit strong vertical stratification. Notably, Deep soil layers (below 20 cm) store over half of global soil organic carbon and serve as key nutrient reservoirs for deep-rooted plants, yet remain understudied relative to surface horizons[16]. Understanding depth-dependent stoichiometry is thus essential for elucidating terrestrial carbon and nutrient cycling and for predicting ecosystem responses to global change in deep soils. Therefore, understanding this depth-dependent variation is essential for interpreting soil functioning and ecosystem responses to disturbance.

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In the context of rapid urbanization, urban forest parks serve as essential "green infrastructure" by providing valuable ecosystem services and supporting sustainable urban development[17,18]. While these parks improve environmental quality and deliver key ecological services, such as runoff reduction, air and water purification, carbon sequestration, and noise mitigation[19], the most distinctive feature of these parks compared to natural forests lies in their soil ecosystems. Urban forest soils, unlike natural forest soils, are shaped by both natural disturbances (e.g., drought, wildfire, storms, insects) and intensive human interventions (e.g., soil compaction, external inputs, irrigation, fertilization)[20]. These "natural-social-economic" coupled systems experience unique patterns of nutrient distribution and cycling. Natural disturbances strongly affect forest biodiversity, directly or indirectly influencing soil properties through changes in vegetation and microbial community structure[21]. Concurrently, the impact of human activities on forest ecosystems can be as intense as that of climate change. For example, human activities such as trampling increase soil bulk density, decrease pH, and lower organic matter, while urbanization alters soil nutrient structures, typically resulting in lower C, N, and P levels in urban than suburban soils[22]. Research in Chengdu's Longquan Mountain Urban Forest Park highlights how soil fertility varies spatially, primarily driven by both natural processes and human reshaping[23]. Thus, the unique dynamics of urban forest soils not only set these ecosystems apart but also make them critical subjects for research into sustainable urban environments.

Numerous studies show that climate, topography, vegetation, and soil conditions influence soil nutrient levels in natural forests[24,25]. Climate (temperature and precipitation) explains about 30% of the variation in soil C, N, and P[26,27]. More rainfall enhances microbial activity and plant nutrient uptake, which promotes soil nutrient cycling[28–30]. In dry climates, microbial activity and vegetation cover decrease, but available soil nutrients can increase[31]. In temperate regions, warming boosts soil N stocks[32]. Higher temperatures at lower elevations accelerate mineralization and the breakdown of organic matter, thereby adding mobile nutrients to the soil. Soil and ecosystem types also strongly affect total C, N, and P[26,27]. Converting primary forest to cropland or grassland reduces soil organic carbon (SOC)[33]. Locally, soil texture and chemistry shape tropical plant communities and mycorrhizal types, affecting nutrient availability[34]. However, these climate–nutrient relationships are based mainly on natural forests. Whether the same patterns occur in urban forest parks is unclear, as the urban heat island effect and anthropogenic activities (e.g., irrigation, soil compaction, fertilization, drainage) may change or override climate-driven soil processes. Most studies focus on natural or agricultural systems rather than urban forests, which are shaped by human management. Urban forest soils are affected by both natural and socio-economic processes[35]. Additionally, soil depth influences responses to various environmental drivers. For instance, the impact of precipitation on nutrient availability typically decreases with increasing soil depth. Surface soils are primarily affected by recent rainfall through enhanced decomposition and leaching, while deeper soils are more influenced by long-term water fluxes and the upward movement of nutrients via roots[28–30]. Therefore, examining the vertical variation of the soil profile is essential for understanding soil ecosystem function in urban environments. Thus, studying the vertical variation of the soil profile is key to understanding soil ecosystem function in urban environments.

Previous studies have shown that spatial variation in soil nutrients results from both natural (e.g., parent material and topography) and anthropogenic (e.g., human activities) factors[20]. Complex interactions among soil nutrient elements influence nutrient bioavailability and ecological function. Specifically, synergistic effects occur when one nutrient enhances the uptake or function of another (such as N and P co-limitation). Conversely, antagonistic effects arise when one nutrient inhibits the uptake of another (e.g., high K^+ reduces Mg^{2+} availability) in soil systems. Stoichiometric ecology provides a framework for understanding these complex interactions among nutrient elements (i.e., elemental ratios such as C/N, C/P, N/P), but its application in urban forest environments remains limited[3,4]. Because urban forest parks are unique ecosystems, how environmental factors drive the spatial heterogeneity of soil nutrients and their stoichiometric ratios, and whether these mechanisms change with soil profile depth, remains largely unexplored. It is important to investigate how environmental drivers of soil nutrient concentrations and stoichiometry vary with soil depth in these unique urban forest parks. This investigation is a necessary first step toward understanding their resilience to global change.

This study selected six forest parks in a single city in southern China as sampling sites. All parks are at elevations below 300 m and share similar soil parent materials. This limits climatic and edaphic variation allowing us to focus on vertical stratification mechanisms under uniform background conditions. We investigated soil profiles from the surface layer (0–10 cm) to the deep layer (40–60 cm). We measured total and available C, N, P, K, and pH as indicators of nutrient status. Using statistical methods (ANOVA, principal component analysis, redundancy analysis, and partial least squares path modeling), we set three objectives. First, we aim to uncover spatial patterns of soil nutrient concentrations and their stoichiometric ratios across parks and soil layers. Second, we seek to quantify the explanatory power of environmental factors (climate, elevation, vegetation, and soil properties) on nutrient scores and stoichiometry. Third, we will identify pathways by which these factors influence soil nutrients and stoichiometry, focuing on whether these pathways differ across soil depths. Our study provides an inference-based understanding of how anthropogenic disturbance may influence vertical nutrient dynamics. These findings aim to inform management practices. Practices include reducing surface compaction, retaining litter, and protecting deep root zones. The goal is to maintain soil health and sustain the ecological functions of urban forests under ongoing urbanization and global change.

Data and methods

Study area

The study was conducted in six forest parks in Shenzhen, a coastal city in southern China (Fig. 1). These parks were established between 1997 and 2005 and have been under active management for approximately 20–27 years, including routine maintenance such as trail construction, irrigation, fertilization, and visitor management. While we do not directly quantify anthropogenic disturbance, we can still gain insights by comparing parks

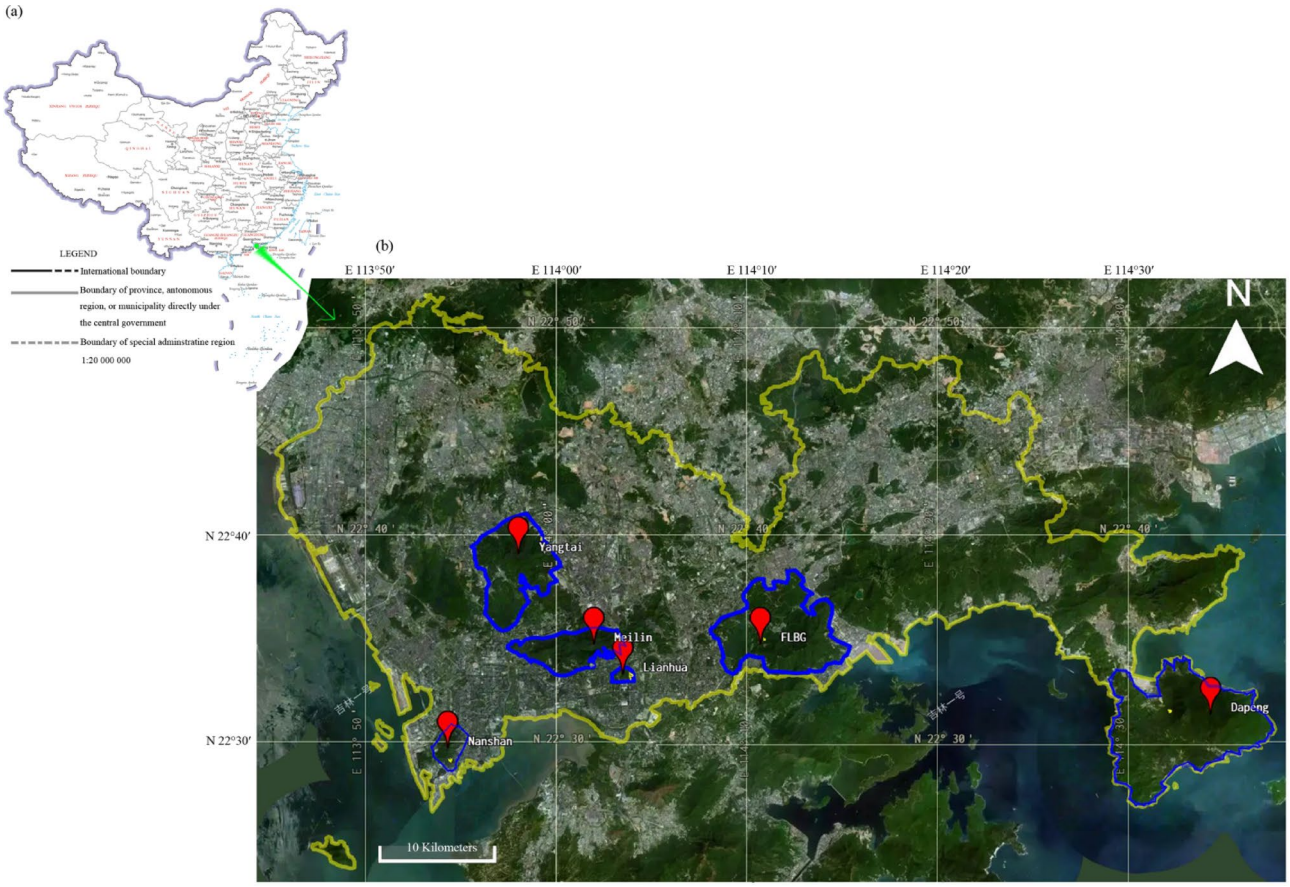


Fig. 1. Map of the study area showing the location of forest parks within Shenzhen, China.

with different management intensities and recreational pressure. Management intensity varies among parks: low-elevation parks (e.g., Lianhua, Meilin) receive higher visitor foot traffic, more frequent irrigation and fertilization, and regular litter removal, whereas high-elevation parks (e.g., Yangtai, Dapeng) are less intensively managed and experience lower recreational pressure. All field investigations, soil sampling, and data analysis conducted in this study were carried out with permission from the Shenzhen Municipal Administration and Comprehensive Law Enforcement Bureau, China, which manages the forest parks involved in this research. The dominant natural soil type is latosolic red soil. Meteorological data indicate an annual average temperature of 23.3 °C, with the lowest monthly average in January (15.7 °C) and the highest in July (29.0 °C). The average annual sunshine duration is 1853.0 h, and the average annual precipitation is 1932.9 mm, 86% of which occurs during the flood season (April to September). Summers, lasting over six months, are dominated by southerly winds, bringing high temperatures and rainfall; autumn and winter are influenced by the northeastern monsoon, resulting in drier conditions. These six forest parks were selected as long-term sample sites, covering diverse geographical locations and elevation gradients to reflect regional environmental heterogeneity. Their geographical and vegetation characteristics are presented in Table 1. According to the Chinese Soil Database, surface soils (0–20 cm) in our study parks are rich in SOC and N; deeper layers (20–60 cm) are moderately rich, but both total and available P and K are poor to extremely poor[36]. All sampled park soils are lateritic red earths from granite or sand-shale weathering. This clay-rich, acidic material develops into highly weathered urban soils with moderate nutrients, medium K, and low P[37,38].

The inset map illustrates Shenzhen's location within China. The base map of China was obtained from the National Standard Map Service of the Ministry of Natural Resources of China (http://bzdt.ch.mnr.gov.cn/index.html; approval No. GS(2019)1686). The Shenzhen study area map was generated using Omap software (version 10.3.2; https://www.ovital.com/) with satellite imagery data sourced from the Jilin-1 Satellite (National 2023), provided by Chang Guang Satellite Technology Co., Ltd. The yellow line denotes the administrative boundary of Shenzhen City. Blue lines represent the boundaries of individual forest parks. The white grid indicates latitude and longitude.

Vegetation survey and soil sampling

To avoid anthropogenic disturbance, sample plots were set up in the interior of each park, away from roads and residential areas. In each park, 3–6 permanent 30 m × 30 m plots were randomly placed. Three nested vegetation quadrats were systematically deployed in each plot. The tree layer used a 20 m × 20 m quadrat, in which species, diameter at breast height (DBH) for individuals > 5 cm, height, and crown width were recorded. The shrub

Park Site	Forest stand	Latitude coordinate	Longitude coordinate	Elevation (m)	Plant community	Soil type
Nanshan Park (Nanshan)	Broad-leaved mixed forest	N22°30′49″–N22°28′46″	E113°53′29″–E113°55′8″	310	*Schima superba; Michelia figo; Corymbia citriodora; Eurya acuminata; Acacia confusa; Schefflera heptaphylla; Aporusa dioica; Litsea rotundifolia var. oblongifolia*	Latosolic red soil
Dapeng Peninsula National Geopark (Dapeng)	Broad-leaved mixed forest	N22°34′4″–N22°26′55″	E114°28′55″–E114°37′33″	842	*Schima superba; Sapium discolor; Cunninghamia lanceolata; Itea chinensis; Aporusa dioica; Litsea rotundifolia* var. oblongifolia	Latosolic red soil
Lianhua Mountain Park (Lianhua)	Broad-leaved mixed forest	N22°33′36″–N22°32′54″	E114°2′56″–E114°4′5″	99	*Ficus hispida; Falcataria moluccana; Corymbia citriodora; Acacia confusa; Schefflera heptaphylla; Aporusa dioica; Psychotria asiatica; Litsea rotundifolia* var. oblongifolia; *Ilex asprella*	Latosolic red soil
Meilin Mountain Park (Meilin)	Broad-leaved mixed forest	N22°35′34″–N22°33′13″	E113°57′28″–E114°3′33″	361	*Acacia confusa; Schima superba; Schefflera heptaphylla; Aporusa dioica; Psychotria asiatica; Ilex asprella; Adina pilulifera*	Latosolic red soil
Fairy Lake Botanical Garden (FLBG)	*Aquilaria sinensis (As)*	N22°38′16″–N22°33′26″	E114°8′8″–E114°15′24″	909	*Aquilaria sinensis; Sapium discolor; Schefflera heptaphylla; Aporusa dioica; Psychotria asiatica; Litsea rotundifolia var. oblongifolia*	Latosolic red soil
	Acacia mangium (Am)					Latosolic red soil
Yangtai Mountain Forest Park (Yangtai)	*Schima superba (Ss)*	N22°40′53″–N22°35′16″	E113°55′39″–E114°0′16″	501	*Schima superba; Castanopsis fissa; Lophostemon confertus; Castanopsis chinensis; Aporusa dioica; Psychotria asiatica; Litsea rotundifolia* var. oblongifolia	Latosolic red soil
	Broad-leaved mixed forest					

Table 1. Geographical information and vegetation characteristics of the sampled plots in the forest parks.

Index	Function	Parameter description
Patrick_index (R)	$R = S$	S is the total number of species in the quadrat
Shannon_Wiener_index (H)	$H = -\sum_{i=1}^{S} P_i \ln P_i$	P_i is the relative importance value of species i, and S is the total number of species.
Margalef's_Richness_Index (Ma)	$Ma = \frac{S-1}{\ln N}$	S is the total number of species in the plot, and N is the total number of individuals of all species
Pielou's_Evenness_Index (J)	$J = \frac{H}{\ln S}$	H is the Shannon–Wiener index, and S is the total number of all species in the quadrat
Simpson's_Diversity_ Index (D)	$D = 1 - \sum_{i=1}^{S} P_i^2$	P_i is the relative importance value of species i, and S is the total number of all species in the quadrat

Table 2. Formulas used for calculating vegetation diversity indices.

layer used a 5 m × 5 m quadrat to record species, height, coverage, and abundance. The herbaceous layer used a 1 m × 1 m quadrat to note species, height, and coverage. Geographical coordinates and elevation for each quadrat were recorded with a GPS device.

In each plot, soil samples were collected from six points in an "S"-shaped pattern. At each point, soil was excavated to a depth of 60 cm, and samples were collected at the following fixed depth intervals: 0–10 cm, 10–20 cm, 20–40 cm, and 40–60 cm. We did not sample by genetic horizons because their thickness varies among parks due to differences in management, vegetation, and disturbance. Instead, we used fixed-depth intervals to ensure site comparability and to assess vertical stratification without confounding variable horizon thickness. Soils from the same horizon at all points were thoroughly mixed to form a composite sample. Visible plant litter, roots, and rocks were removed, with unweighed roots noted as a limitation. The composite samples were air-dried, homogenized, and passed through a 2 mm sieve (< 10-mesh) before physicochemical analysis.

Plant diversity indices

Alpha-diversity indices (see Table 2 for a list) were calculated based on vegetation survey data following Nicholas and Anne[39]. Specifically, the Patrick index (R), Shannon-Wiener index (H), Margalef's Richness Index (Ma), Pielou's Evenness Index (J), and Simpson's Diversity Index (D) were included. These indices were derived from the comprehensive importance value of each species by integrating relative abundance, frequency, and coverage.

Soil physicochemical analyses

Soil physicochemical properties were determined using standard methods: Soil pH was measured potentiometrically with a water-to-soil ratio of 2.5:1. Soil organic carbon (SOC) was determined by the potassium dichromate oxidation-spectrophotometric method. This method is widely recognized as a reliable approach for routine soil SOC analysis, although it may slightly underestimate SOC relative to dry combustion techniques[40]. All samples were processed under identical conditions, using the same standardized protocol, thereby ensuring robust, internally valid comparisons within this study. We did not apply a correction factor to our SOC values, as our primary objective was to facilitate internal comparisons among parks and soil depths, rather than to provide absolute quantification suitable for cross-study meta-analyses. Previous studies have employed correction factors (typically ranging from 1.05 to 1.32) to address the slight underestimation of SOC by the dichromate method compared with dry combustion techniques[40]. Total nitrogen (TN) was measured

using the Kjeldahl method; available nitrogen (AN) was determined by the alkali hydrolysis-diffusion method. Total phosphorus (TP) was determined by the alkali fusion-molybdenum antimony anti-spectrophotometric method; available phosphorus (AP) was extracted with sodium bicarbonate and measured by the molybdenum antimony anti-spectrophotometric method. Total potassium (TK) was measured by the sodium hydroxide fusion-flame photometry method; and available potassium (AK) was extracted with ammonium acetate and determined by atomic absorption spectrophotometry.

Environmental variables

Automated weather stations (CR1000X, Campbell Scientific, Inc.) installed in each park continuously recorded meteorological data throughout both the sampling year and the previous year. Parameters extracted for analysis included peak temperatures (PTemp), air temperature (Ta), soil temperature (Ts), relative humidity (RH), soil volumetric water content (VWC), and soil electrical conductivity (EC). Annual averages or cumulative values for these parameters were used in subsequent analyses. For brevity in figures and tables, the following abbreviations are used: PTemp (peak temperature), Ta (air temperature), Ts (soil temperature), RH (relative humidity), VWC (soil volumetric water content), and EC (soil electrical conductivity).

Statistical analyses

Data normality and homogeneity of variances were checked using Shapiro–Wilk and Levene's tests, respectively. Data were log-transformed when necessary to meet ANOVA assumptions. Soil samples collected at varying depths from the same core were treated as repeated measures nested within each sampling location (plot). To account for this inherent non-independence, linear mixed models (LMMs) with restricted maximum-likelihood (REML) estimation were utilized. Soil depth (0–10, 10–20, 20–40, 40–60 cm) was modeled as a fixed categorical factor, and sampling location (plot) was included as a random intercept. The effect of depth was assessed using Satterthwaite's approximation for degrees of freedom. Pairwise post hoc comparisons among depth levels were performed with Tukey adjustment using the emmeans package.

Next, principal component analysis (PCA) was performed on standardized soil nutrient data to reduce dimensionality and derive a comprehensive soil nutrient score for each sample based on the first two principal components (PCs). The relationships among soil nutrient indicators, stoichiometric ratios, and environmental factors were then examined using Spearman's rank correlation analysis. Subsequently, redundancy analysis (RDA) was applied to quantify the explanatory power of environmental variables for the observed variation in soil nutrient and stoichiometric concentrations. As climate and topographic variables remain constant within individual parks, RDA results pertaining to these variables are presented descriptively, and significance tests (p-values) are not reported.

Building on these analyses, a partial least squares path model (PLS-PM) was constructed to elucidate the direct and indirect effects of vegetation attributes, geographical factors, and climate conditions on soil nutrients and their stoichiometric ratios. In structuring the PLS-PM, five latent variables were defined, each using specific observed variables. Topography was represented by elevation. Climate comprised peak temperature (PTemp), air temperature (Ta), soil temperature (Ts), relative humidity (RH), soil volumetric water content (VWC), annual average precipitation (AAP), and soil electrical conductivity (EC). Plant diversity was assessed using species richness (R), Shannon-Wiener index (H), Margalef index (Ma), Pielou index (J), and Simpson index (D). For soil properties, observed variables included soil organic carbon (SOC), total nitrogen (TN), total phosphorus (TP), total potassium (TK), pH, alkaline hydrolyzable nitrogen (AN), available phosphorus (AP), and available potassium (AK). Lastly, stoichiometric ratios included C/N, C/P, N/P, and AN/AP. All variables were standardized before analysis. All variables were standardized before analysis. Model quality was assessed using goodness-of-fit (GoF) indices. As climate, elevation, and topography are constant within each park, the effective sample size for park-level variables is limited to six, rendering statistical tests underpowered. Accordingly, only descriptive effect sizes (explained variances and path coefficients) are reported for these variables. In contrast, within-park analyses utilize the complete dataset and are presented with significance tests.

All statistical analyses and graphical outputs were produced using R software (version 4.2.0; R Core Team, 2022; https://www.r-project.org/) in conjunction with the lme4, lmerTest, emmeans, vegan, plspm, and ggplot2 packages. Statistical significance for within-park comparisons (e.g., depth effects) was set at $p < 0.05$; no significance was attributed to park-level drivers. Data were organized, and figures were edited using WPS Office (version 12.2.0; https://www.wps.com/).

Results

Analysis of soil nutrient content and ecological stoichiometry characteristics

Vertical distribution and inter-park differences in soil total nutrients, pH and available nutrients

LMMs that accounted for repeated measures indicated that both SOC and TN declined significantly with increasing soil depth. The 0–10 cm layer contained significantly higher SOC and TN values than all deeper layers. TP also exhibited a significant depth effect: concentrations in the 0–10 cm layer exceeded those in the 20–40 and 40–60 cm layers, though the difference between the 0–10 and 10–20 cm layers was not significant. In contrast, TK did not display a significant trend with depth. In Meilin Park, TP was descriptively higher in the surface layer than in deeper layers (Fig. 2a–h, Tables S1 and S2).

Regarding soil pH, values also varied notably: soils in Lianhua Park were near-neutral (pH 5.72–5.93), whereas all other parks were strongly acidic (pH < 4.3). Soil pH exhibited significant variation with depth. Post hoc comparisons indicated that the 0–10 cm layer had a higher pH than the 10–20 cm and 20–40 cm layers, but was slightly lower than the 40–60 cm layer (Fig. 2e, Tables S1 and S2).

Among parks, Lianhua had significantly higher SOC, TN, and TK levels overall. Dapeng exhibited a significantly higher surface SOC content than all other parks. Nanshan and Yangtai showed clear vertical

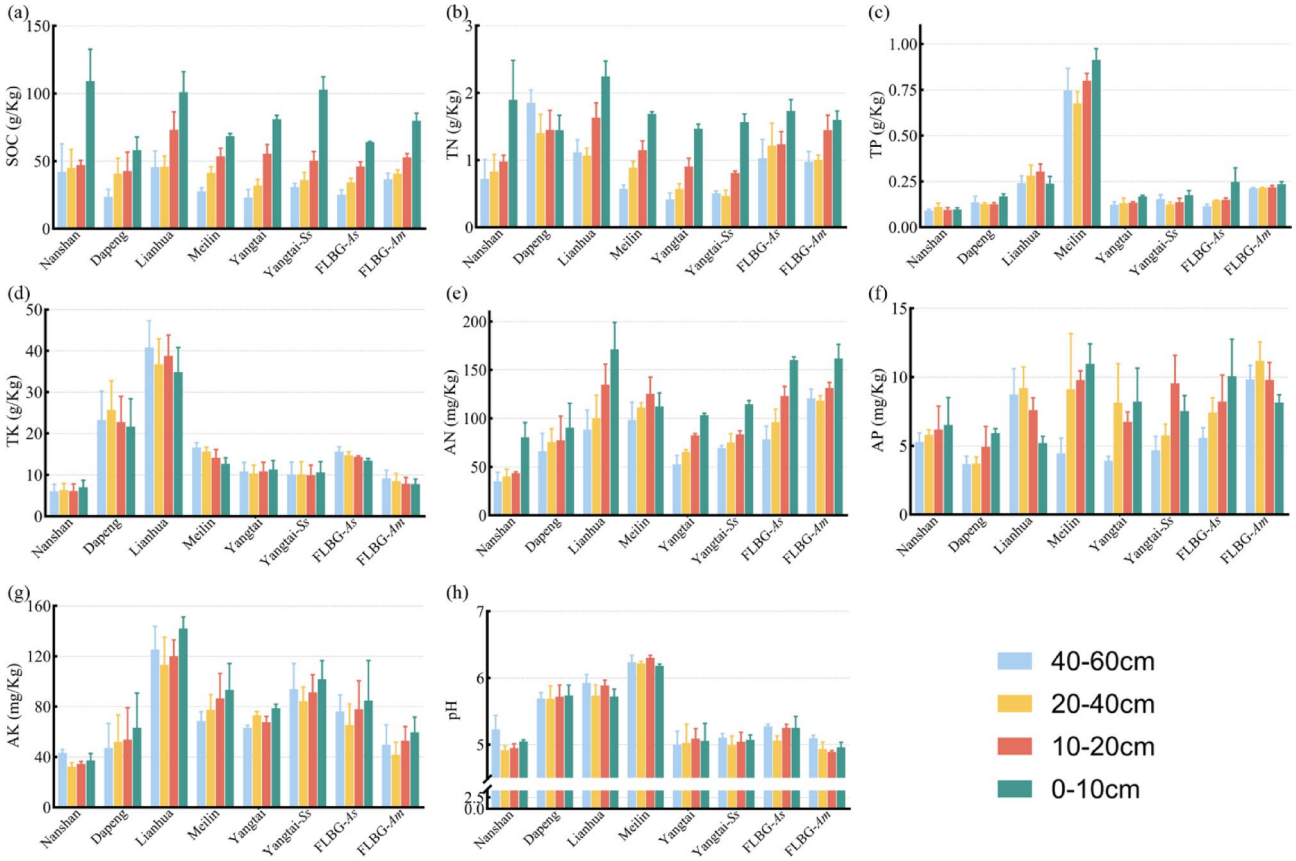


Fig. 2. Distribution of soil nutrients across soil profiles in different forest parks. SOC: soil organic carbon (**a**); TN: total nitrogen (**b**); TP: total phosphorus (**c**); TK: total potassium (**d**); AN: alkaline hydrolyzable nitrogen (**e**); AP: available phosphorus (**f**); AK: available potassium (**g**); pH (**h**). Bars show means, and error bars show standard errors. Sample sizes: Lianhua Park, n = 6 for all depths; all other parks, n = 3 for each depth. Depth effects were assessed using linear mixed models (LMMs) with sampling point as a random intercept; post-hoc comparisons were performed with Tukey adjustment. Detailed statistical results (F-statistics, degrees of freedom, p-values, and post-hoc comparisons) are provided in Supplementary Tables S1 and S2.

differentiation in SOC and TN in certain layers. Meilin had the highest TP across all depths, and TK was highest in Lianhua and lowest in Nanshan and FLBG-Am. TK also varied significantly between forest types within the same park (Fig. 2a–h, Table S1).

AN decreased markedly with increasing soil depth, while AK also showed a significant depth effect. In contrast, AP did not demonstrate a statistically significant overall depth effect. Notably, AP values tended to be higher in the 10–20 cm or 20–40 cm layers in some parks (e.g., Lianhua, Meilin), whereas surface values were relatively elevated in others (e.g., Yangtai, FLBG-As). However, these observations are descriptive and should be interpreted with caution, given substantial spatial variability (Fig. 2i–n, Tables S1 and S2).

Soil ecological stoichiometry

Stoichiometric ratios showed differences in nutrient balance and deficiency. The C/N ratio was highest in Yangtai (Yangtai Science Park; Yangtai-Ss) and Nanshan Park, and lowest in Dapeng Park. The C/N ratio exhibited a significant depth effect; however, post hoc tests revealed no significant pairwise differences among individual depth levels, indicating that the overall trend was weak. In contrast, both C/P and N/P ratios decreased with increasing depth, with the 0–10 cm layer displaying significantly higher values than deeper layers. AN/AP did not vary significantly with depth; consequently, inter-park differences in AN/AP are presented descriptively in Table S1 and Fig. 3d. Nanshan had the highest surface C/P and N/P ratios, suggesting a greater potential for P deficiency (Fig. 3b, c, Table S1).

Relationship between environmental factors and soil nutrients

PCA result

PCA based on 12 soil nutrient indices extracted five principal components, with a cumulative variance contribution rate of 87.12%, sufficiently representing the original nutrient information (Fig. 4a, Table S3). Based on this analysis, a composite fertility score was derived from Z-score standardized data and PCA loadings. The mean value for pure stands was lower than for broadleaf mixed forests, but the difference was not statistically significant (Fig. 4b). In contrast, scores among parks decreased in the order: Lianhua, Dapeng, FLBG, Meilin,

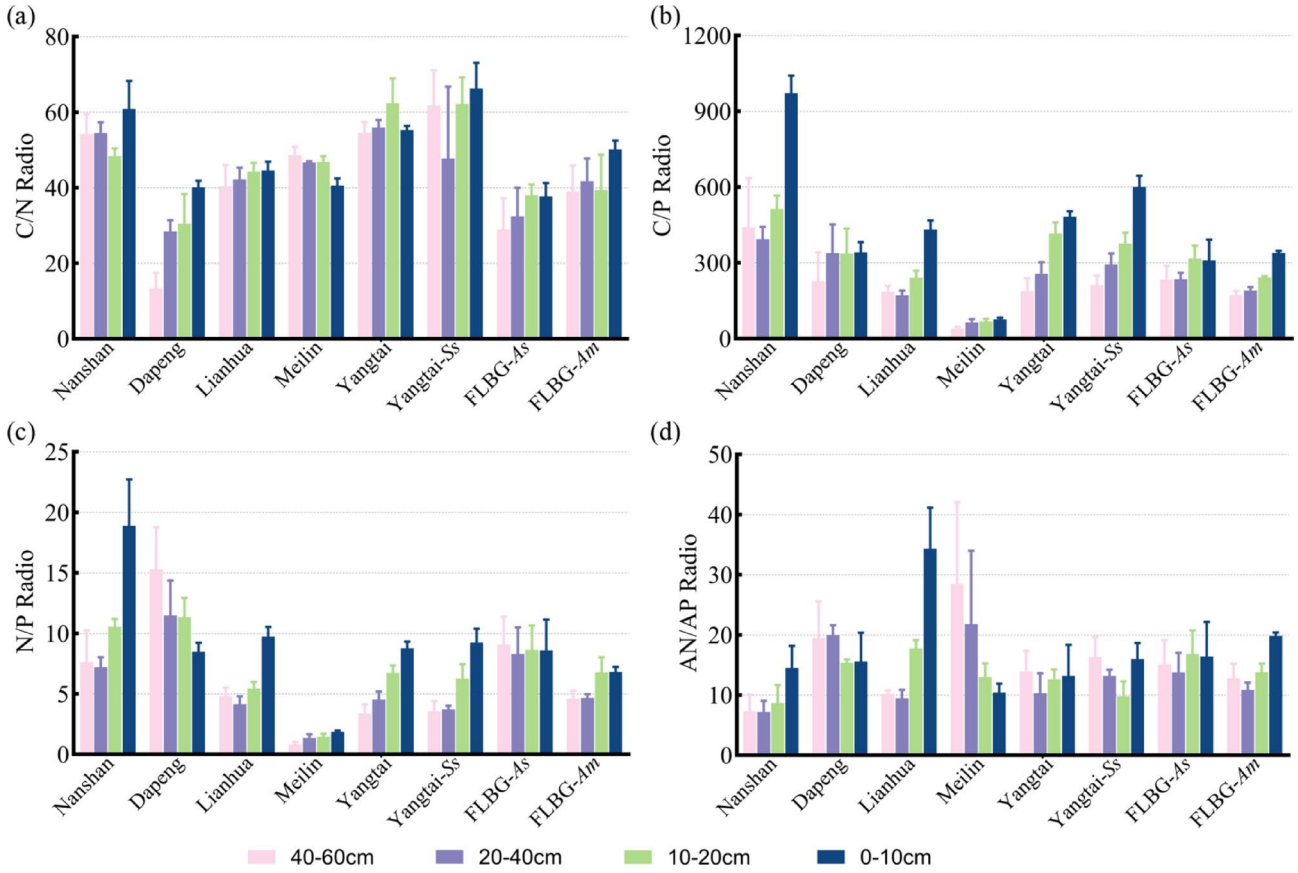


Fig. 3. Distribution of soil nutrient stoichiometric ratios across soil profiles in different forest parks. (**a**) C/N Ratio; (**b**) C/P Ratio; (**c**) N/P Ratio; (**d**) AN/AP Ratio. Bars show means, and error bars show standard errors. Sample sizes: Lianhua Park, n = 6 for all depths; all other parks, n = 3 for each depth. Depth effects were assessed using linear mixed models (LMMs) with sampling point as a random intercept; post-hoc comparisons were performed with Tukey adjustment. Detailed statistical results (F-statistics, degrees of freedom, p-values, and post-hoc comparisons) are provided in Supplementary Tables S1 and S2.

Nanshan, and Yangtai (Fig. 4c). The score declined with increasing elevation (Fig. 4d). Increased annual precipitation significantly raised the score (Fig. 4e). The score also declined with increasing soil depth, with the 0–10 cm and 10–20 cm layers having significantly higher scores than deeper layers (Fig. 4f).

Correlation and ordination analysis results

Correlation and ordination analyses were conducted to examine relationships among soil nutrients, stoichiometric ratios, and environmental variables. Two relationships were particularly noteworthy: the C/P ratio was positively correlated with TP, suggesting that phosphorus accumulation drives variation in C/P; and the AN/AP ratio was positively correlated with both AN and AP, indicating that this ratio reflects coupled nitrogen and phosphorus availability (Fig. 5a, b). Soil depth exhibited a negative correlation with SOC, TN, and several other nutrient metrics, corroborating the depth-profile results. Additional correlation results, including those involving pH, temperature, humidity, elevation, and park-level comparisons, are provided in the Supplementary Materials (Table S3; Fig. S3).

RDA further clarified the influence of environmental factors on soil nutrients and stoichiometry. As climate, elevation, and forest stand type remain constant within each of the six parks, the following percentages are provided as descriptive effect sizes, and no significance tests are reported. AAP, VWC, and Ts explained 20.4%, 18.0%, and 19.7% of the variance in soil nutrients, respectively, showing positive associations with TN and TK and negative correlations with TP, AP, and AK (Fig. 6a, c). Plant diversity indices (H) each explained less than 10% of variance and were positively correlated with AP, AK, and TP, but negatively with TN and TK. Forest stand type and EC explained 11.0% and 19.7%, respectively, and were positively associated with SOC but negatively with TP and AN. Soil depth explained 16.1% of variance and was negatively correlated with SOC, AK, and AP, and positively with TP, TN, AN, and TK (Fig. 6a, c, and Table S4).

For stoichiometric ratios, AAP and PTemp explained 18.3% and 15.0% of variance, respectively, and were positively correlated with N/P and AN/AP but negatively with C/N (Fig. 6b, d). Forest stand, EC, and RH explained 9.5%, 19.1%, and 16.3%, respectively, and were positively associated with C/N and C/P but negatively with AN/AP. Elevation explained 14.5% of variance and correlated positively with C/N and negatively with C/P and N/P (Fig. 6b, d). Vegetation diversity indices accounted for only a minor proportion of stoichiometric

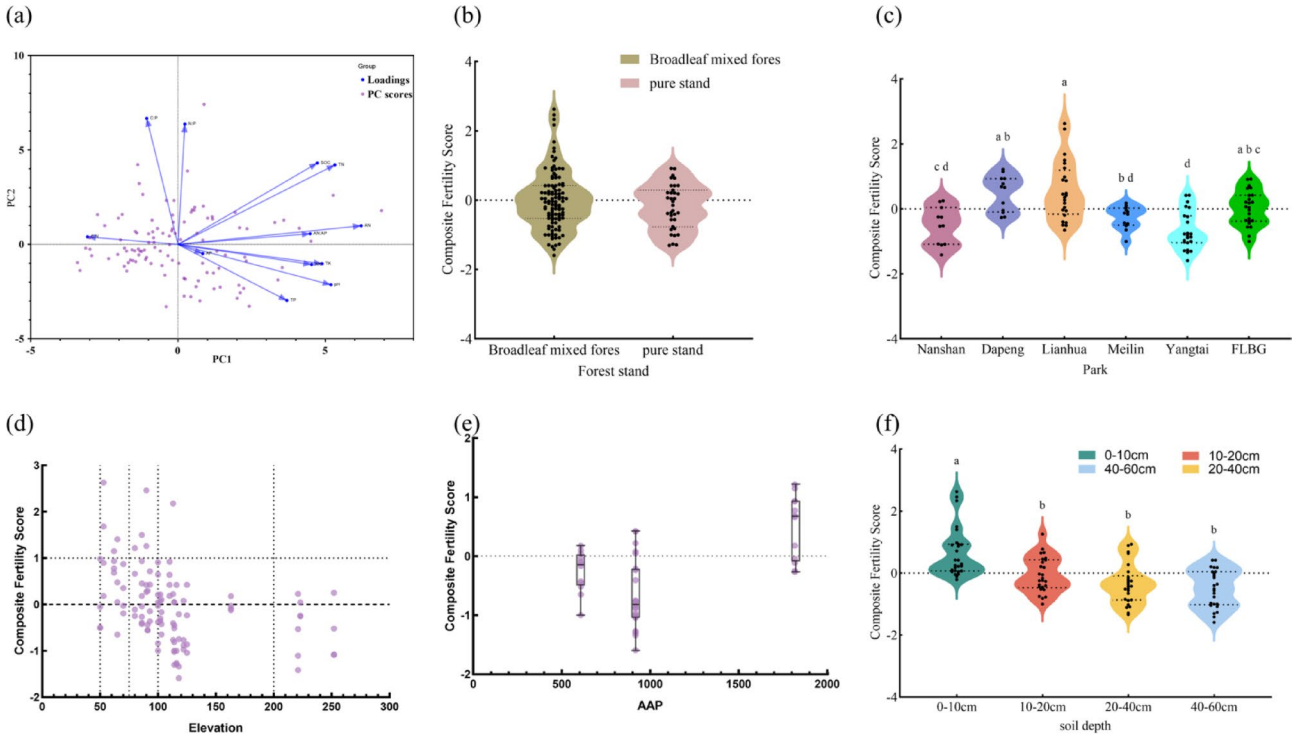


Fig. 4. Comprehensive assessment of soil nutrient status. (**a**) Principal component analysis (PCA) of soil nutrients. (**b**) Composite fertility score by forest stand type. (**c**) Composite fertility score across parks. (**d**) Fertility score versus elevation. (**e**) Fertility score versus annual precipitation (AAP). (**f**) Fertility score by soil depth. Panels (**b**), (**c**), and (**f**) were analyzed using one-way ANOVA with Duncan's test. Different lowercase letters denote significant differences at $p < 0.05$.

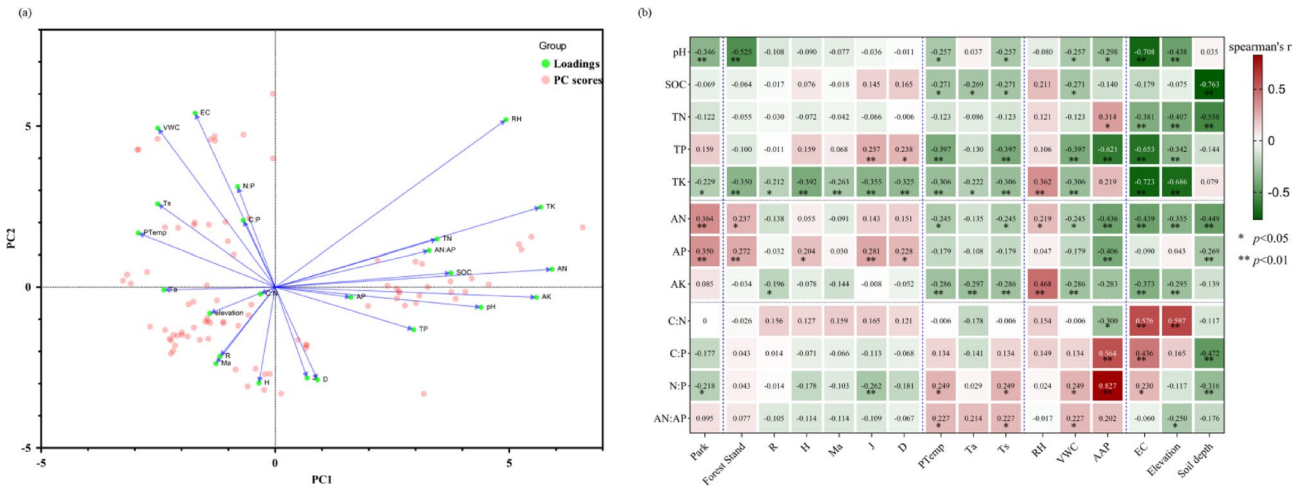


Fig. 5. Relationships between environmental factors and soil nutrients. (**a**) Principal component analysis (PCA) biplot of environmental factors, soil nutrients, and their stoichiometries. (**b**) Correlation matrix between environmental factors and soil nutrients/stoichiometries. Red and green cells indicate positive and negative correlations, respectively, with numerical r values and significance levels indicated: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

variation (Table S5). As these environmental variables are correlated, the marginal explained percentages sum to > 100% and do not represent independent or additive contributions. To clarify unique versus shared variance, we performed variance partitioning (Table S6), which revealed that climate explained 31.8% (nutrients) and 28.4% (ratios) of the variation uniquely, whereas vegetation explained 25.4% and 14.2%, soil depth explained 12.3% and 5.2%, and topography explained 5.9% and 13.2%. The total explained variance (adjusted R^2) was 60.1%

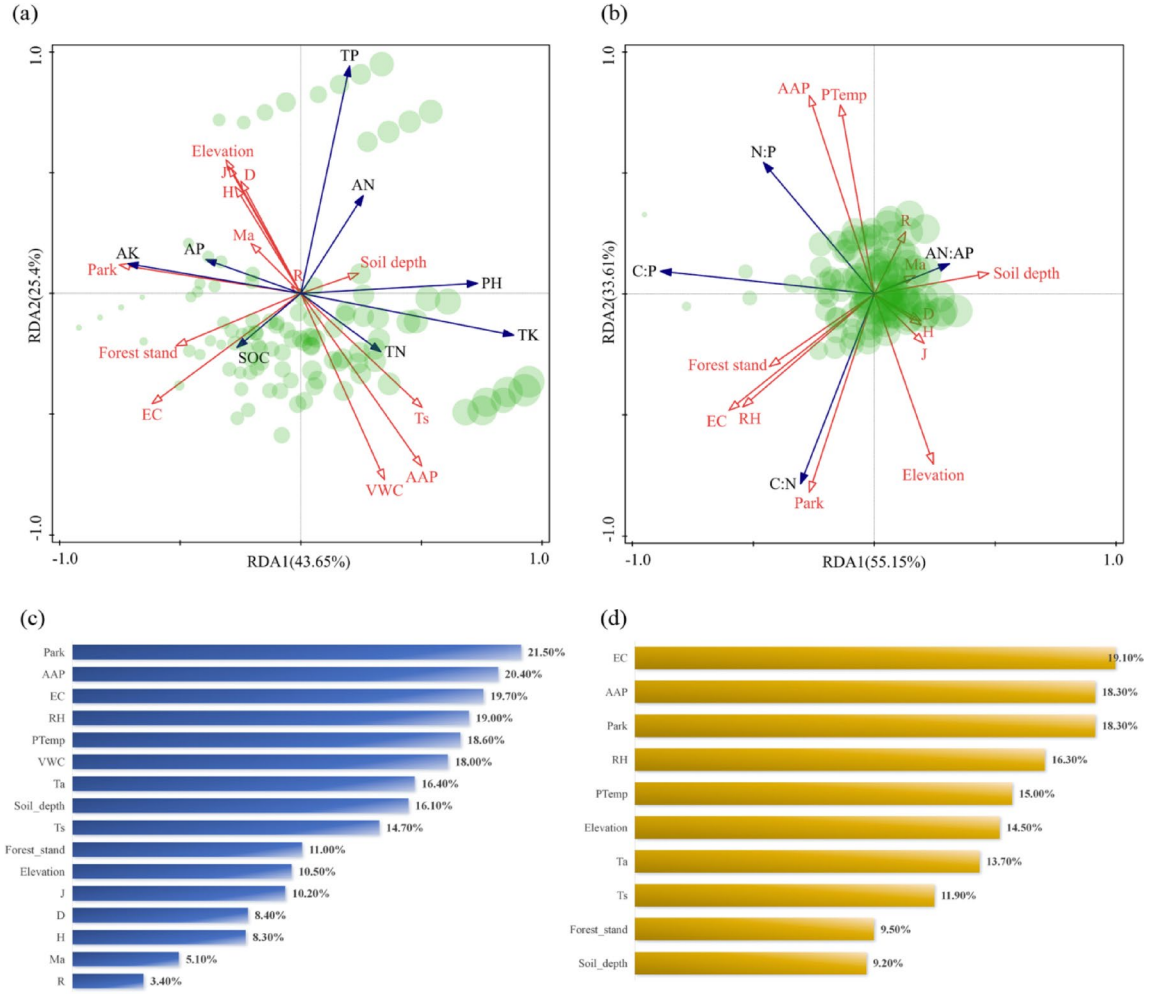


Fig. 6. Effects of environmental variables on soil nutrients and their stoichiometries. (**a**) Redundancy Analysis (RDA) triplot showing the influence of environmental factors (red arrows) on soil nutrients (blue arrows). (**b**) RDA triplot showing the influence of environmental factors (red arrows) on soil nutrient stoichiometries (blue arrows). (**c**, **d**) Pie charts displaying the relative contributions (%) of each environmental factor to the variances in (**c**) C, N, P contents and (**d**) their stoichiometric ratios. Variance partitioning presents unique contributions (adjusted R^2) of predictor groups. No significance tests are reported for park-level groups due to the limited sample size (n = 6).

for nutrients and 48.2% for ratios (Table S6). Given the limited number of parks (n = 6) for park-level variables (climate, elevation, and forest stand type), these variance partitioning results are presented descriptively and should be regarded as exploratory.

Mechanisms driving environmental effects on soil nutrients and stoichiometry

PLS-PM analysis revealed how topography, climate, and plant diversity shape soil properties and stoichiometric ratios. These influences vary with soil depth (0–20 cm vs. 20–60 cm). Topography showed consistently negative direct associations with soil properties (i.e., SOC, TN, TP, TK, AN, AK, AP, and pH; see Fig. 6) in both layers, as well as indirect associations via climate. Climate demonstrated a negative association with soil properties, although this relationship diminished with increasing depth. In contrast, the effect of soil properties on stoichiometry shifted from negative at the surface (− 0.695) to positive in the deeper layer (0.663). Plant diversity was weakly positively associated with surface stoichiometry (0.150) but strongly negatively associated at depth (− 0.464), suggesting depth-dependent limiting factors (Fig. 6a, b). The negative effect at depth may mean greater plant diversity intensifies nutrient competition or alters root-associated microbes, reducing stoichiometric ratios, while surface effects are influenced by anthropogenic inputs and litter mixing. The model fit indices (GOF) were 0.436 for the surface layer and 0.397 for the deeper layer. The explanatory power (R^2) for latent variables was at least 0.47 at both depths (Fig. 7a, b). Taken together, these results demonstrate that environmental effects on soil stoichiometry vary with depth: topography and climate are the primary influences in surface soils, whereas plant diversity and internal soil properties are more influential in deeper layers. Given the small sample size of parks (n = 6) for park-level variables (climate, topography), the PLS-PM results are reported descriptively and should be regarded as exploratory, rather than confirmatory.

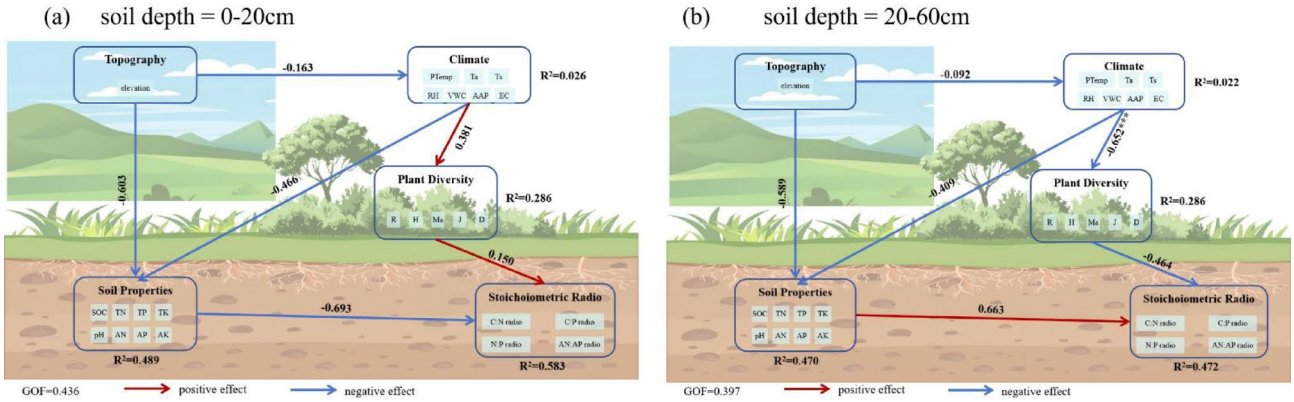


Fig. 7. Results of the partial least squares path modeling (PLS-PM) analysis, illustrating the direct and indirect effects of environmental factors on soil nutrients and their stoichiometries in (**a**) topsoil (0–20 cm) and (**b**) subsoil (20–60 cm).

Solid arrows represent significant path coefficients, with arrow width proportional to the strength of the effect. Red and blue colors indicate positive and negative relationships, respectively. GOF: goodness-of-fit statistic for the entire PLS-PM model. The coefficients of determination (R^2) for each endogenous latent variable are shown, representing the proportion of variance explained by the model. Path coefficients are presented descriptively. As climate and topography are constant within parks, bootstrap confidence intervals and p-values are not reported. The model is considered exploratory. The climate and soil element icons were sourced from Vecteezy.com (Free License; https://www.vecteezy.com/).

Discussion

Spatial heterogeneity of soil nutrients and their stoichiometry in urban forest parks

This study aimed to (1) characterize the vertical distribution of soil carbon, nitrogen, and phosphorus stoichiometry in six urban forest parks within a single Chinese city, all below 300 m elevation and with similar parent material; (2) assess how environmental factors such as climate, topography, vegetation, and soil properties explain nutrient variation; and (3) evaluate whether the main drivers of nutrient dynamics differ by soil depth.

The P deficiency, reflected in elevated C/P, N/P, and AN/AP ratios (see Study area, Methods), is a consequence of the lateritic red-earth parent material and is further modulated by anthropogenic disturbance. Anthropogenic disturbance has not erased this parent-material-driven P deficiency, but it has altered its vertical expression. In natural forests, P typically declines with depth[41,42]. In contrast, in our study parks, available phosphorus (AP) peaks in the 10–40 cm layer rather than at the surface. This pattern may emerge because soil sealing and compaction, combined with regular litter removal, which depletes the surface organic P pool, restrict surface P availability[43–45]. In high-visitation parks such as Lianhua and Meilin, surface soil compaction from recreational trampling is plausible. Meanwhile, downward transport of soluble P allows subsequent root-induced solubilization in deeper layers[38,46]. Fine roots, often concentrated in the 10–30 cm range in compacted urban soils, release previously fixed P through rhizosphere processes such as organic acid exudation and phosphatase activity[47,48]. Historical deep fertilization may also contribute to this distribution[49]. Therefore, the subsurface AP maximum results directly from interactions between human management practices and soil mineralogy, distinguishing urban park soils from natural forest soils.

Similarly, SOC and N decrease with depth, matching natural trends[9,10], but the decrease is modulated by management[47,50]. Surface soils are richer in oxygen, organic matter, and moisture, promoting microbial and root activity, which leads to more carbon and nutrient accumulation than in deeper layers[11,43,44,51]. Most roots are near the surface, maximizing water and nutrient uptake, while consistent inputs from litter and roots outweigh leaching losses[12,13,52]. Deeper soils, by contrast, receive less organic input and retain fewer leached nutrients. As a result, leaching does not increase N or P in deeper layers in these conditions[14,15]. Additionally, soil EC explained a significant portion of stoichiometric ratio variation. EC, reflecting soluble salt concentration, affects nutrient availability and microbial activity and was positively associated with C/N and C/P ratios in this study[53]. Higher EC might reduce decomposition efficiency or shift plant–microbe nutrient partitioning, and could signal the accumulation of base cations that compete with P, influencing its availability[53]. These findings suggest EC, often overlooked, is a potentially important indicator of disturbance and ionic stress in urban soils. In summary, while parent material sets the baseline for P and K deficiency, human activities create unique depth profiles (e.g., subsurface AP peak) and introduce novel drivers (EC) that are absent in natural forests.

Key environmental factors driving soil nutrient variation

Elevation showed a negative correlation with total soil nutrients (at all sites < 300 m), contrary to the positive elevation-nutrient trend in many natural ecosystems[54]. This hypothetical explanation for the reversal is that local anthropogenic pressure, rather than elevation itself, is the cause. Low-elevation parks (Lianhua, Meilin) are nearer the city center and experience higher visitor trampling, more irrigation and fertilization, and regular litter removal, all of which disrupt nutrient accumulation. Higher-elevation parks (Yangtai, Dapeng)

are more remote, with less management and recreational pressure. Although we did not quantify management intensity, this interpretation aligns with studies reporting negative elevation-nutrient relationships in disturbed landscapes[54]. Future research should systematically quantify management-related variables, such as foot traffic and maintenance frequency, to directly assess their influence on soil nutrient patterns.

Climate also drives nutrient variation: AAP explained 20.4% of nutrient variance, Ts 19.7%, and soil VWC 18.0%. While these values are comparable to those in natural forests[55–57], in urban environments, the direction and magnitude of climate effects are altered by human activities. For example, higher Ts were negatively correlated with nutrient content, likely because urban heat island effects accelerate decomposition beyond natural rates[58]. Similarly, although moderate precipitation enhances nutrient availability[28,59], excessive rainfall can increase erosion and leaching[60], and these processes are more pronounced in urban parks due to compacted soils and disrupted hydrology[26,27,61,62]. Thus, climate-driven patterns in urban areas often reflect the modifying influence of anthropogenic disturbance, rather than natural processes alone.

Plant diversity explained little of the nutrient variation (< 10%), which contrasts with the strong diversity-nutrient relationships typical of natural ecosystems[63–65]. In urban parks, this weak relationship is mainly due to human actions: vegetation is planted or managed, and anthropogenic activities (such as irrigation, fertilization, and trampling) frequently disrupt natural plant-soil processes[66,67]. Plant diversity in these forest parks has a dual origin: pure stands were planted, while mixed forests regenerated naturally with later management. Thus, diversity and nutrient patterns result from both planting and ecological processes, possibly explaining weak or unexpected patterns observed. Urban plant diversity also stems from both deliberately planted pure stands and naturally regenerated mixed forests, so 'forest type' is a more meaningful predictor than species richness. Forest type explained 11% of nutrient variation and 9.5% of stoichiometric variation, positively correlating with SOC and C/N. Overall, these findings indicate that anthropogenic disturbance and abiotic stress are likely dominant factors shaping urban soil nutrient patterns[68].

Vertical differentiation of soil ecological mechanisms in urban forest parks

The main drivers of soil stoichiometry shift with depth: in surface layers (0–20 cm), climate, topography, and human management are dominant; in deeper layers (20–60 cm), plant diversity and inherent soil properties become more influential. This depth-dependent transition stands in contrast to many natural forests, where biological controls often prevail even in surface horizons[69]. The surface layer is where human activities and natural processes interact closely, and management practices often obscure natural nutrient dynamics. Specifically, management practices (fertilization, irrigation, trampling) interact strongly with topography and climate, while the effects of plant diversity are suppressed[66,67]. Consequently, plant diversity shows a weak positive association with surface stoichiometry. However, in deeper layers, it exhibits a strong negative association. Although this negative association is counterintuitive, it can be explained: in deeper, less-disturbed soils, plants compete more intensely for limited nutrients (especially P), and diverse root systems may exacerbate competition or alter microbial communities, thereby reducing stoichiometric ratios[54].

The 20–40 cm soil layer can be regarded as a 'self-sustaining layer', meaning that its nutrient cycling and supply depend primarily on internal biological processes (e.g., root activity, mycorrhizal associations, microbial decomposition, and necromass turnover) rather than on external anthropogenic inputs such as fertilization, irrigation, or litter removal. As human management practices rarely reach this depth, plants must rely on deep root architecture and mycorrhizal symbiosis to explore and modify their environment, thereby influencing the activation and turnover of deep soil nutrients[69]. Concurrently, inherent soil properties (e.g., cation exchange capacity, stability of organic-mineral complexes) replace external inputs as key factors maintaining nutrient supply, explaining why "soil properties" transition from a suppressive to a promoting factor in deeper layers.

Overall, sustainable management of urban forests often focuses on short-term soil fertility. However, it is also important to emphasize protecting the self-sustaining capacity of deep soil layers (20–60 cm). This can be achieved by minimizing deep disturbance, planting deep-rooted native species, and promoting litter return. Future research should directly quantify management intensity and routinely use soil EC as an indicator of urban disturbance. This study has several limitations. First, because soil sampling occurred only once, our results capture a single point in time and may not reflect seasonal variation. Future research should include multiple seasons to distinguish temporal from spatial effects. Second, we did not assess microbial properties such as biomass, community structure, or enzyme activity, so our interpretations of microbial processes are speculative and require direct validation. Third, anthropogenic disturbance was inferred from park location and elevation rather than measured directly. Additionally, with climate and topographic variables held constant within each park, our park-level analyses are limited by a small sample size (n = 6), making these results exploratory. The most robust findings are those comparing depth profiles within parks. Due to the study's focus on a single low-elevation city with uniform parent material, our conclusions are most applicable to similar urban forest settings and should not be broadly generalized without further evidence.

Conclusion

This study shows that different factors control soil C, N, and P at various depths in subtropical forest parks. Phosphorus deficiency, as indicated by high C/P and N/P ratios, is closely linked to soil acidification and leaching. Surface soils (0–20 cm) are shaped mainly by climate and topography, while deeper layers (20–60 cm) reflect soil properties and plant processes, showing soil development and biological adaptation. Human activities disrupt natural nutrient balances and weaken the link between plant diversity and soil fertility. These results highlight the need to consider vertical differences in driving factors when modeling landscape biogeochemistry. To better understand ecosystem resilience, include both surface and deep-soil processes and their interactions with landscape features.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Received: 4 February 2026; Accepted: 16 July 2026
Published online: 20 July 2026

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Acknowledgements

We acknowledge the Soil SubCenter, National Earth System Science Data Center, National Science & Technology Infrastructure of China (http://soil.geodata.cn) for providing data support on soil classification standards in natural ecosystems.

Author contributions

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Funding

This work was supported by the Shenzhen Science and Technology Program (grant numbers: KCXST20221021111211025, SYSPG20241211173609007); and the Scientific Foundation of the Urban Management Bureau of Shenzhen (grant numbers: 202018 & 202401).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-63150-6.

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